

EDUCATION

Lovely Professional University
Bachelor of Technology (Computer Science & Engineering)
Jalandhar, India
July 2022 - May 2026

SKILLS SUMMARY

- **Programming:** Python, SQL, JavaScript, TypeScript
- **AI & ML:** Machine Learning, Deep Learning, Generative AI, LLMs, LLM APIs, AI Agents, Prompt Engineering, Model Evaluation
- **NLP & RAG:** NLP, RAG, Embeddings, Qdrant, Vector Databases, Semantic Search, LangChain, LangGraph
- **Geospatial:** GeoJSON, GeoTIFF, Computer Vision, Geospatial Data Processing
- **Backend:** FastAPI, REST APIs, PostgreSQL, JSON
- **Engineering:** Testing, Debugging, Performance Optimization, Git
- **Infrastructure:** AWS, Docker, CI/CD, Monitoring

WORK EXPERIENCE

- AI Intern | Know AI | [LINK](#) June 2026 - August 2026
- Built Generative AI and LLM applications using Python, focusing on prompt engineering, experimentation, and evaluation.
 - Developed and tested AI workflows and REST APIs using LLMs, APIs, tool calling, automation, and RAG.
 - Designed **RAG pipelines** using embeddings, vector databases, and semantic search for knowledge-grounded retrieval.
 - Developed LLM and agent evaluation pipelines measuring accuracy, reliability, tool performance, and output quality.
 - Improved AI reliability through monitoring, troubleshooting, debugging, latency optimization, and failure handling.
- Data Engineer Intern | flickDONE | [LINK](#) June 2025 - August 2025
- Built scalable Python AI/ML data pipelines processing 50,000+ records for preprocessing and feature engineering.
 - Engineered automated **data preprocessing and validation workflows**, reducing manual verification by 70%.
 - Optimized AWS-based AI/ML pipelines, improving execution speed by 25% through performance tuning.
 - Developed reusable AI workflows for document processing, LLM applications, API integrations, and enterprise automation.
 - Collaborated with teams to evaluate AI/ML approaches and deliver scalable, reliable workflows.

PROJECTS

- Ledgerly — Multi-Tenant SaaS Billing Platform | [LIVE](#) April 2026 - September 2026
- Built a full-stack SaaS platform using Next.js, NestJS, PostgreSQL, Docker, REST APIs, and testing.
 - Implemented tenant-isolated customer and invoice workflows with **RBAC and transactional data operations**.
 - Built **Stripe payment workflows and webhook processing** with idempotency and transactional updates.
 - Added **rate limiting, health checks, and secure error handling** for production-grade reliability.
 - Developed **174 automated tests across 21 suites** covering authentication, billing, payments, and tenant isolation.
- Production-Grade RAG Engine | [LIVE](#) June 2024 - August 2024
- Built an end-to-end production-grade RAG system using Python, FastAPI, LangChain, Qdrant, and AWS.
 - Developed document preprocessing, chunking, embeddings, indexing, retrieval, and response generation pipelines.
 - Implemented **semantic vector search with Qdrant** for low-latency, knowledge-grounded context retrieval.
 - Added reranking and retrieval evaluation to improve retrieval quality, context relevance, and AI response performance.
 - Improved AI service reliability through testing, monitoring, fallback handling, and error analysis.
- Geospatial Deep Learning Segmentation | [LIVE](#) August 2025
- Built an end-to-end deep learning segmentation pipeline using Python, PyTorch, U-Net, and computer vision techniques.
 - Developed image preprocessing and annotation workflows for satellite imagery datasets and segmentation masks.
 - Trained and evaluated deep learning models for pixel-level building and road segmentation from geospatial imagery.
 - Built inference pipelines and API workflows for processing images and generating reliable segmentation prediction outputs.
 - Generated GeoJSON and GeoTIFF outputs for infrastructure mapping and geospatial analysis.

CERTIFICATES

- Python, Data Science & ML (Cipherschools) | [CERTIFICATE](#) August 2024
- Built machine learning models using PyTorch and applied ML techniques for model development and evaluation.
 - Developed **Dockerized environments** supporting reproducible experimentation and deployment
- Generative AI with Large Language Models (Coursera) | [CERTIFICATE](#) March 2024
- Applied prompt engineering techniques for production LLM applications.
 - Built **retrieval workflows, AI output evaluation, and RAG pipeline** performance validation.